Linear equations use one (or more) variables where one variable is dependent on the other(s).

The longer the distance we travel by a taxi, the more we have to pay. The distance travelled is an example of an *independent variable*. Being dependent on the distance, the taxi fare is called the *dependent variable*.

The more one uses electricity, greater will be the amount of electricity bill. The amount of electricity consumed is an example for independent variable and the bill amount is naturally the dependent variable.

3.10.2 Graph of a linear function in two variables

We have talked about parallel lines, intersecting lines etc., in geometry but never actually looked at how far apart they were, or where they were .Drawing graphs helps us place lines. A linear equation is an equation with two variables (like x and y) whose graph is a line. To graph a linear equation, we need to have at least two points, but it is usually safe to use more than two points. (Why?) When choosing points, it would be nice to include both positive and negative values as well as zero. There is a **unique line** passing through any pair of points.

Example 3.44

Draw the graph of $y = 5x$

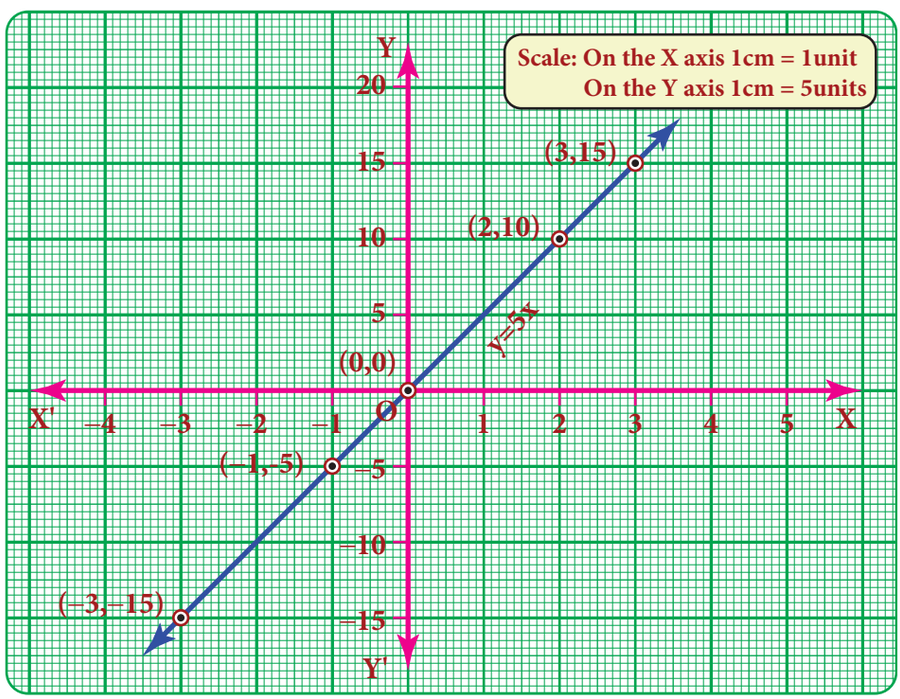
x	−3	−1	0	2	3
y	−15	−5	0	10	15

x	$y = 5x$
−3	$y = 5 \times (-3) = -15$
−1	$y = 5 \times (-1) = -5$
0	$y = 5 \times (0) = 0$
2	$y = 5 \times (2) = 10$
3	$y = 5 \times (3) = 15$

Solution:

The given equation $y = 5x$ means that for any value of x, y takes five times of x value.

Plot the point (−3,−15) (−1,−5) (0,0) (2,10) (3,15)



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